

Ivan Gerardo

Senior AI/ML Engineer

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Profile

Senior AI/ML Engineer with 10+ years of experience designing, developing, and deploying production-grade machine learning, deep learning, and large language model solutions. Strong background in building scalable AI systems, cloud-based data platforms, and end-to-end ML pipelines across financial services, e-commerce, SaaS, and enterprise environments.

Formulated and refined LLM applications using Python, TensorFlow, and Hugging Face, boosting inference speeds by 40% and improving end-user response times.

Orchestrated end-to-end machine-learning pipeline across AWS, GCP, and Azure, handling 3 TB daily, cutting data latency by 40 minutes and eliminating three major ingestion failures.

Skills & Abilities

- **AI/ML & Deep Learning:** Python, PyTorch, TensorFlow, scikit-learn, XGBoost, CNNs, LSTMs, Transformers, NumPy, Pandas
- **Generative AI & LLMs:** OpenAI, Hugging Face, LangChain, RAG, Fine-Tuning, Prompt Engineering, AI Agents, Pinecone, FAISS
- **MLOps & Cloud:** MLflow, Kubeflow, Airflow, SageMaker, Vertex AI, GitHub Actions, CI/CD, AWS, GCP, Azure
- **Data & Analytics:** Snowflake, BigQuery, PostgreSQL, Feature Engineering, A/B Testing, NLP, Computer Vision, OCR
- **Software & Deployment:** FastAPI, gRPC, Microservices, Docker, Kubernetes, IaC, Automated Testing, Real-Time Inference

Experience

SENIOR AI/ML ENGINEER | SPARKCOGNITION (US) | JULY 2023 – PRESENT

- Established robust monitoring frameworks for AI models trained on AWS SageMaker, reducing drift by 25% and improving overall model robustness metrics.
- Engineered a multi-modal deepfake verification pipeline that fuses Vision Transformers (ViT), Wav2Vec2, and CLIP embeddings to deliver synchronized face-voice authentication with industry-leading accuracy.
- Delivered real-time detection at 35 FPS by integrating OpenCV, FFmpeg, and TensorRT for GPU-accelerated feature extraction and optimized inference.
- Designed a hybrid ensemble classifier (CNN + Transformer fusion) that achieved 97.9% accuracy on in-the-wild deepfake datasets, outperforming baseline and academic benchmarks.
- Championed the integration of Vertex AI Pipelines and MLflow for model retraining and regression testing, shortening deployment lifecycles by an average of fifteen days.
- Orchestrated the creation of end-to-end predictive pipelines for retail, employing LSTM and XGBoost to forecast demand and optimize supply chain strategies, reducing stock-outs by 15%.
- Integrated critical sales, inventory, and promotion data from 5+ sources into BigQuery daily, shrinking data latency from 24 hours to less than 30 minutes.
- Designed and implemented interactive Tableau dashboards and automated alerting mechanisms, leading to a 15% reduction in stock-out and overstock incidents and validating significant operational return.
- Instituted an AI document intelligence system that extracts, classifies, and summarizes high-volume legal/financial documents, achieving a 70% reduction in compliance manual review time.

- Devised and trained 12 custom Transformer models, each achieving >92% extraction accuracy on test sets, directly boosting document processing reliability.
- Spearheaded the establishment of comprehensive CI/CD pipelines with Terraform and GitHub Actions, accelerating the release cycle for ML artifacts by 30%.
- Created API-driven automation microservices using Zapier, n8n, and Make.com, orchestrating workflows spanning data ingestion, async notifications, and AI inference.
- Introduced robust microservice inference APIs that handled high-volume traffic at low latency, decreasing critical system failures by 35% and enhancing visibility.

AI/ML OPS ENGINEER | THOUGHT MACHINE (UK) | OCTOBER 2020 – JUNE 2023

- Transformed ML pipeline infrastructure across AWS, GCP, and Azure for AI-driven banking, reducing pipeline deployment time by 3 days and enhancing scalability by 45%.
- Initiated and finalized predictive models for fraud detection and personalization, boosting overall model accuracy by 32% and reducing false positive alerts by 24%.
- Launched API-first ML microservices using FastAPI, gRPC, and Node.js, delivering predictions to core banking and frontend applications with sub-second response times.
- Automated model training, deployment, and monitoring through CI/CD pipelines using GitHub Actions, Terraform, Docker, Kubernetes, and Helm, reducing ML deployment time by 60% and improving release reliability across production environments.
- Implemented MLOps observability using Datadog, Prometheus, Grafana, and ELK Stack to monitor model performance, feature drift, pipeline health, and SLA compliance, reducing production incident response time by 35%.
- Integrated multi-modal AI systems combining structured transaction data, OCR-based financial documents, PDFs, and NLP-driven customer interactions, improving data extraction efficiency by 40% and enabling faster risk and compliance analysis.
- Accelerated ML inference workflows, delivering sub-second predictions and increasing transaction risk scoring performance by 50% on massive financial data streams.

FULL STACK ENGINEER | BIGCOMMERCE (US) | MARCH 2017 – SEPTEMBER 2020

- Directed the development and refinement of ETL workflows in C# and T-SQL, merging large e-commerce data into structured models, enhancing reporting pipeline efficiency by 45%.
- Orchestrated the deployment of automated reporting solutions and RESTful APIs, utilizing FastAPI, Flask, SQL, and T-SQL to resolve the three biggest causes of reporting delays.
- Integrated data analytics and machine learning models into the core SaaS platform, enabling merchants to monitor revenue, conversions, and customer behavior in real time and improving decision-making speed by 35%.
- Advanced frontend analytics dashboards using React, Redux, and Chart.js, amplifying data visualization, interactivity, and KPI tracking accuracy for key stakeholders across three distinct e-commerce sites.
- Strengthened system responsiveness by refining database queries and indexing; cut dashboard and API response times by 40%, bolstering scalability for future growth.
- Played a key role in migrating legacy PHP monoliths to a modular service-oriented architecture using Docker, AWS ECS, and microservices best practices, reducing deployment complexity by 50%.
- Partnered with cross-functional teams to identify data gaps, refine business requirements, and align reporting solutions with business goals, compliance standards, and long-term product strategy.

Education

B.S. in COMPUTER SCIENCE | 2014 - 2017 | University of Chester